State Estimation in Electric Power Systems Using PMU Data

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***Abstract*—State estimation enables the internal variables of electric power systems to be monitored from available measurements. The integration of Phasor Measurement Units (PMUs) has led to hybrid methods that provide greater accuracy than approaches based solely on Supervisory Control and Data Acquisition (SCADA) measurements. This paper reviews the fundamentals of state estimation, the statistical treatment of measurement errors, and the main estimation methods, with particular emphasis on the role of hybrid techniques in modern power-system monitoring.**

***Index Terms*—State estimation, PMU, SCADA, electric power systems.**

I. Introduction

A. State Estimation

Throughout the development of electric power systems, critical events such as blackouts and service interruptions have motivated the creation of increasingly robust operation and control tools. The growing complexity of interconnected systems created a need for real-time monitoring, leading to the implementation of SCADA systems capable of collecting data from remote terminal units (RTUs). However, measurement errors, communication failures, and incorrect configurations may prevent the acquired data from accurately representing the actual system state. State estimation was developed to address these limitations. It is now an essential function of Energy Management Systems (EMSs), providing a reliable database for the safe and efficient operation of the grid [1, 2].

State estimation optimally determines the voltage magnitudes and phase angles at the system buses from redundant and often imperfect measurements. The process is based on statistical criteria, most commonly the minimization of the sum of the squared differences between measured and estimated values [8]. Its implementation smooths random errors, detects grossly erroneous measurements, and helps compensate for missing data. These capabilities are essential for maintaining the quality of the information used in critical applications such as economic dispatch, security analysis, and voltage regulation [4].

The measurements most commonly available in a power system are [3]:

- Voltage magnitudes: Voltage magnitudes are measured using voltmeters with a specified accuracy. A voltage-magnitude measurement is available at nearly every system bus i.
- Active- and reactive-power flows: Active-power-flow measurements are generally available at both ends of each transmission line. Such a measurement may represent the active power flowing from bus i to bus j through line $i \to j$. Reactive-power-flow measurements are similarly available at both ends of many transmission lines.
- Active- and reactive-power injections: The active-power injection at a bus i may also be measured, as may the corresponding reactive-power injection.

As shown in Fig. 1, information concerning the estimated state and topology changes is processed by an internal topology-processing module. An observability analyzer then determines whether the available measurements provide complete system coverage. Next, a prefilter removes clearly invalid measurements. The estimation block computes the optimal system-state values, although its results remain vulnerable to errors in data, topology, or network parameters. A subsequent block therefore identifies and interprets these errors from the statistical characteristics of the estimation results [9]. The

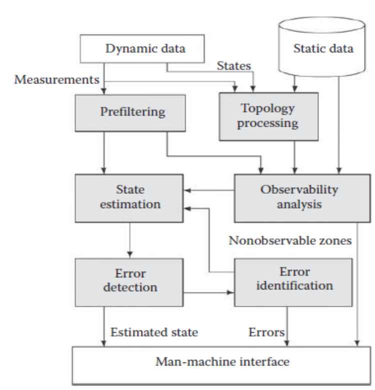

Fig. 1. Block diagram of the state-estimation process. Adapted from [1].

operational value of state estimation is evident in its use within energy control centers, where it accurately determines the system operating point and supports contingency studies, expansion planning, reliability assessment, and real-time decision-making. Since its introduction, state estimation has become a cornerstone of modern monitoring and control systems.

It supports contingency and reliability analyses, expansion planning, security-constrained economic dispatch, and voltage regulation while ensuring that the analytical model faithfully represents actual grid conditions [2, 4].

B. PMUs

The growing demand for greater temporal resolution and accuracy in power-system monitoring has encouraged the use of Phasor Measurement Units (PMUs) to complement conventional SCADA measurements. Whereas SCADA typically provides data every few seconds without precise time synchronization, PMUs record synchronized voltage and current phasors at reporting rates commonly ranging from 30 to 120 samples per second [1, 2]. Synchronization is achieved through GPS signals that provide a common time reference with microsecond-level accuracy. Measurements from different substations can therefore be compared directly without separately estimating their relative phase shifts [2]. PMUs consequently provide a substantial improvement in system observability and dynamic-analysis capability.

A PMU comprises several functional blocks: analog inputs from instrument transformers, anti-aliasing filters, GPS-synchronized analog-to-digital converters, and a microprocessor that calculates phasors using techniques such as the Discrete Fourier Transform (DFT). This architecture produces highly accurate synchronized phasors, which are aggregated by Phasor Data Concentrators (PDCs) and transmitted to control centers [1, 2]. IEEE Std C37.118 specifies requirements for the accuracy and consistency of these measurements. Despite their high resolution, PMUs remain subject to errors arising from instrumentation channels and system imbalance; their limitations must therefore be clearly defined [1]. Figure 2 presents a simplified diagram of PMU operation.

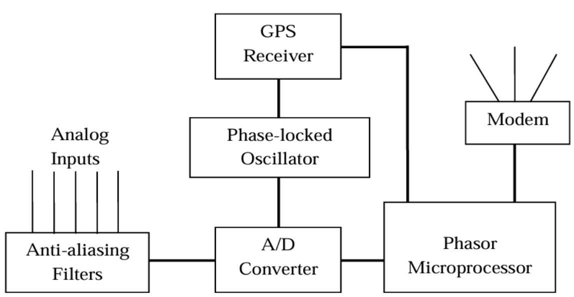

Fig. 2. Operation of a generic PMU. Adapted from [2].

In this diagram, PMUs are installed at power-system substations and provide time-stamped positive-sequence voltage and current measurements for monitored buses and feeders, together with frequency and rate-of-change-of-frequency measurements. The data are stored locally, although local storage capacity is limited. Devices at the next hierarchical level are commonly known as PDCs. They collect data from several PMUs, discard invalid data, align time stamps, and create a coherent synchronized record covering a larger region of the power system. PDCs also provide local storage.

The integration of PMUs into state estimators has been shown to improve system robustness, reliability, and observability. Major benefits include better detection of topology errors, fewer required critical measurements, and greater sensitivity to measurement errors at nearby substations. PMU data from neighboring networks may also supplement local information. Fully realizing these benefits requires strategic PMU placement and the use of both voltage and current phasors. Despite their potential, the high cost of large-scale deployment makes installation at every bus of a large system impractical. PMUs are therefore generally installed at selected substations according to the intended application, and a substantial body of research focuses specifically on optimal PMU placement [11]. Most power systems consequently operate with hybrid schemes that combine SCADA and PMU data, requiring estimation algorithms that can accommodate different reporting rates and measurement types [1, 8].

Because PMU deployment is limited and installations are often geographically dispersed, an appropriate architecture integrating PMUs, communication links, and data concentrators is required to maximize the benefits of a synchrophasor measurement system. Figure 3 shows an example of a widely accepted architecture.

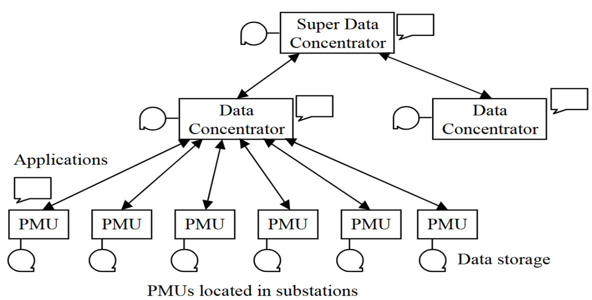

Fig. 3. Architecture of a PMU-based system. Adapted from [11].

By recording measurements in near real time, PMUs have facilitated the transition from static state estimation (SSE) to more advanced approaches such as dynamic state estimation (DSE), which captures system transitions and behavior following changes in load or generation. This development reflects the fact that actual power systems do not remain strictly stationary; conventional SSE techniques may therefore be unable to represent their behavior adequately [9]. PMU measurements provide the foundation for dynamic algorithms that track system variations in real time.

II. State-Estimation Methods

State estimation seeks to determine the complex voltage phasor at each bus of a power system; these phasors constitute the system state [1]. The estimator processes measurements obtained from acquisition and monitoring systems such as SCADA and PMUs. These may include active- and reactive-power flows, current and voltage magnitudes at buses and along lines, and the operating status of circuit breakers, tap-changing transformers, shunt capacitors, and reactors.

Topological constraints and network parameters, including transmission-line and transformer impedances, are also considered [6].

Measurements differ in location, type, and accuracy according to the system under consideration. In many cases, they are supplemented with pseudo-measurements—values inferred from historical data—and structural network constraints to improve redundancy in areas of low observability [1]. Reliable state estimation is essential before security studies or operational optimization can be performed because direct measurements, such as nodal power injections, may be unavailable or contain significant errors [6].

The problem has traditionally been addressed using Weighted Least Squares (WLS), which formulates a nonlinear nodal model and solves it iteratively until a consistent power-system operating point is obtained [4]. Numerical ill-conditioning, however, has motivated alternatives based on constrained normal equations, orthogonal transformations, pseudoinverse techniques such as the Peters–Wilkinson method, and hybrid or augmented-matrix formulations [5].

Robust techniques such as Weighted Least Absolute Value (WLAV) estimation and the Schweppe–Huber Generalized M-estimator (SHGM) have also been proposed. Nevertheless, WLS remains the method most widely used in practice because of its computational efficiency and comparatively low implementation cost in real-world control centers [9].

Understanding these methods requires a clear mathematical formulation. In (1), z is the measurement vector; $h(x)$ is the vector of nonlinear functions relating the system state to measurements such as voltages, power flows, and active- and reactive-power injections; x is the vector of state variables to be estimated; and e is the measurement-error vector [2, 3].

$$\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_m \end{bmatrix} = \begin{bmatrix} h_1(x_1, \cdots, x_n) \\ h_2(x_1, \cdots, x_n) \\ \vdots \\ h_m(x_1, \cdots, x_n) \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_m \end{bmatrix} = \mathbf{H}(\mathbf{x}) + \mathbf{e} \quad (1)$$

- Active- and reactive-power injections

$$P_i = V_i \sum_{j=1}^{n} V_j \left(G_{ij} \cos(\theta_{ij}) + B_{ij} \sin(\theta_{ij}) \right) \quad (2)$$

$$Q_i = V_i \sum_{j=1}^{n} V_j \left(G_{ij} \sin(\theta_{ij}) - B_{ij} \cos(\theta_{ij}) \right) \quad (3)$$

- Active- and reactive-power flows

$$P_{ij} = V_i^2 \left(G_{s,i} + G_{ij} \right) - V_i V_j \left(G_{ij} \cos\theta_{ij} + B_{ij} \sin\theta_{ij} \right) \quad (4)$$

$$Q_{ij} = -V_i^2 \left(B_{s,i} + B_{ij} \right) - V_i V_j \left(G_{ij} \sin\theta_{ij} - B_{ij} \cos\theta_{ij} \right) \quad (5)$$

For a system with N buses, the state vector x has the following form when bus 1 is selected as the reference bus:

$$x^T = [\theta_2 \ \theta_3 \ \theta_N \ V_2 \ V_3 \ V_N]$$

The following assumptions are commonly made regarding the statistical properties of measurement errors, although they are not always fully justified:

- The errors follow a normal distribution.
- The expected value of every error is zero.
- The errors are independent.

The following subsections present several state-estimation methods based on this formulation.

A. Weighted Least Squares

Weighted Least Squares (WLS) is one of the most common methods for solving the state-estimation problem [6, 1, 15, 4, 2]. It determines the state that minimizes a weighted sum of squared measurement errors. Starting from (1), the error vector can be written compactly as

$$\mathbf{e} = \mathbf{z} - \mathbf{H}(\mathbf{x})$$

This expression represents the difference between the measurements and their true but unknown values, which are approximated using the estimated state variables $\check{\mathbf{x}}$. The difference between the actual and estimated measurements can therefore be written as

$$\check{\mathbf{e}} = \mathbf{z} - \check{\mathbf{z}} = \mathbf{z} - \mathbf{H}\check{\mathbf{x}} = \mathbf{e} - \mathbf{H}(\check{\mathbf{x}} - \mathbf{x}) \quad (6)$$

Minimizing the algebraic sum of the errors is unsuitable because positive and negative errors may cancel one another. WLS instead minimizes their weighted sum of squares. Because measurements are affected by errors and disturbances, measurements considered more accurate are assigned greater weight. The resulting objective function is

$$f = J = \sum_{j=1}^{N} w_j e_j^2 \quad (7)$$

The optimal estimate minimizes this function; consequently, the derivative of f, evaluated at the estimated state $\check{x}$, must equal zero:

$$\left. \frac{\partial f}{\partial x_i} \right|_{\check{\mathbf{x}}} = 0$$

After expansion, the true errors are replaced by their estimates. The resulting expression comprises the Jacobian matrix, the diagonal weighting matrix, and the estimated-error vector. The partial derivatives of the error vector $\mathbf{z} - \mathbf{H}\mathbf{x}$ are obtained from those of $\mathbf{H}\mathbf{x}$; when evaluated at the estimated state, they are represented by the elements of $\mathbf{H}$. In matrix form,

$$[\mathbf{H}][\mathbf{W}][\check{\mathbf{e}}] = 0 \quad (8)$$

Substituting (6) into (8) yields (9).

$$\mathbf{H^T W \check{e}} = \mathbf{H^T W} \left(\mathbf{z} - \mathbf{H}\check{\mathbf{x}} \right) = \mathbf{0} \quad (9)$$

Carrying out the matrix operations and solving for the estimated state ($\check{\mathbf{x}}$) gives (10).

$$\check{\mathbf{x}} = \left(\mathbf{H^T W H} \right)^{-1} \mathbf{H^T W z} = \mathbf{G^{-1} H^T W z} \quad (10)$$

This equation introduces the gain matrix ($\mathbf{G} = \mathbf{H^T W H}$). To relate the estimate to the true state, (1) is substituted into (10) as follows:

$$\check{\mathbf{x}} = \mathbf{G^{-1} H^T W (H x + e)} = \mathbf{G^{-1} G x} + \mathbf{G^{-1} H^T W e}$$

After simplification, the state-correction vector is

$$\mathbf{\Delta x} = \check{\mathbf{x}} - \mathbf{x} = \mathbf{G^{-1}H^TWe} \tag{11}$$

The problem is solved iteratively until a stopping criterion is met or the correction becomes sufficiently small. At each iteration, the state estimate is updated according to (12).

$$\mathbf{x^{(k+1)}} = \mathbf{x^{(k)}} + \mathbf{\Delta x^{(k)}} \tag{12}$$

Figure 4 summarizes the algorithm in a block diagram.

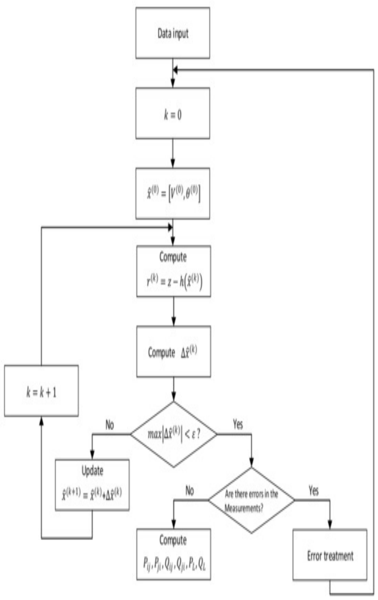

Fig. 4. Block diagram of the WLS state-estimation algorithm. Adapted from [1].

B. Orthogonal Transformation

One of the main limitations of WLS estimation is the numerical difficulty that may arise when the measurement covariance matrix is inverted, particularly when the problem is ill-conditioned or contains elements with extremely different magnitudes. The system matrix may then become nearly singular, reducing estimator stability and accuracy. Orthogonal-decomposition methods, including QR decomposition and the Gram–Schmidt procedure, have been proposed to address these difficulties.

These techniques avoid directly forming or inverting problematic normal-equation matrices and substantially improve the numerical stability of state estimation, particularly when highly accurate constraints or measurements with negligible errors are included. The WLS objective function may be transformed to eliminate R^{-1} as follows [1, 5, 7, 13]:

$$\mathbf{J(\Delta x)} = [\mathbf{\Delta z} - \mathbf{H\Delta x}]^{\mathbf{T}}\, \mathbf{W}\, [\mathbf{\Delta z} - \mathbf{H\Delta x}]$$

$$\mathbf{J(\Delta x)} = \left[\mathbf{\Delta\tilde{z}} - \mathbf{\tilde{H}\Delta x}\right]^{\mathbf{T}} \left[\mathbf{\Delta\tilde{z}} - \mathbf{\tilde{H}\Delta x}\right]$$

This gives

$$\mathbf{J(\Delta x)} = ||\mathbf{\Delta\tilde{z}} - \mathbf{\tilde{H}\Delta x}||^{\mathbf{2}} \tag{13}$$

where

$$\mathbf{\tilde{H}} = \mathbf{W^{\frac{1}{2}}H}$$

$$\mathbf{\Delta\tilde{z}} = \mathbf{W^{\frac{1}{2}}\Delta z}$$

The orthogonal-decomposition algorithm seeks a matrix $\mathbf{Q}$ such that

$$\mathbf{\tilde{H}} = \mathbf{QR} \tag{14}$$

where $\mathbf{Q}$ is an $m \times m$ orthogonal matrix satisfying $\mathbf{Q^T} = \mathbf{Q^{-1}}$, and R is an $m \times n$ trapezoidal matrix of the form

$$R = \begin{bmatrix} U \\ 0 \end{bmatrix}$$

In the absence of round-off error, U is the triangular matrix associated with the factorization of G.

The orthogonality of $\mathbf{Q}$ means that $\mathbf{Q^TQ}$ equals the identity matrix and, consequently, its transpose is equal to its inverse. The matrix $\mathbf{U}$ has an upper-triangular structure; because $\mathbf{H}$ need not be square, the displayed factor may be rectangular. Thus,

$$\mathbf{\tilde{H}} = \mathbf{QU} = \begin{bmatrix} q_{11} & q_{12} & q_{1j} \\ q_{21} & q_{22} & q_{2j} \\ q_{i1} & q_{i2} & q_{ij} \end{bmatrix} \begin{bmatrix} u_{11} & u_{1j} \\ 0 & u_{2j} \\ 0 & 0 \end{bmatrix}$$

Substitution into the least-squares solution gives

$$\mathbf{x^{est}} = \left[U^TQ^TQU\right]^{-1} \left[U^T\right] \left[Q^T\right] \mathbf{\tilde{z}}$$

or, equivalently,

$$\mathbf{x^{est}} = \left[U^TU\right]^{-1} U^T\mathbf{\tilde{z}}$$

Alternatively, the analysis in [1] yields

$$\mathbf{U\Delta x} = \mathbf{Q_n^T\Delta\tilde{z}}$$

where Q_n contains the first n columns of Q. The calculation of Δx at each iteration using orthogonal transformations therefore consists of the following steps:

1) Compute the orthogonal factorization $\mathbf{\tilde{H}} = \mathbf{QR}$.
2) Calculate the right-hand-side vector $\mathbf{\Delta z_q} = \mathbf{Q_n^T\Delta\tilde{z}}$.
3) Obtain $\mathbf{\Delta x}$ by back-substitution in $\mathbf{U\Delta x} = \mathbf{\Delta z_q}$.

The gain matrix G does not need to be formed explicitly. Because of the way the pivots are computed, orthogonal factorization is numerically more robust than its LU counterpart, an important advantage for ill-conditioned systems. It also tolerates the simultaneous presence of very large and very small measurement weights more effectively. Its principal disadvantage is the computational cost of obtaining the orthogonal matrix Q.

C. Kalman Filtering

Before the introduction of PMUs, online state estimation in electric power systems relied on low-rate, unsynchronized SCADA data and was therefore inadequate for precise monitoring of fast system behavior [14]. The increasing deployment of PMUs has made real-time state estimation technically feasible and enabled faster, more accurate methodologies.

This context creates a need for dynamic estimation techniques that are robust, precise, computationally efficient, and capable of producing reliable results with low latency [10]. Dynamic state-estimation methods include Kalman filters, robust dynamic techniques, square-root filtering methods, and artificial-intelligence-based approaches. Kalman-filter variants are widely used because they are comparatively straightforward to implement, can predict future states, and effectively filter measurement noise. Their performance may nevertheless deteriorate in the presence of outliers, motivating the development of more resilient alternatives.

1) Extended Kalman Filter (EKF) Algorithm: The EKF is an efficient recursive algorithm for state estimation in nonlinear systems. It seeks to minimize the covariance of the estimation error between the actual and estimated states [14, 10]. A general nonlinear stochastic discrete-time model and its measurement equation can be expressed as

$$\begin{aligned} x_{k+1} &= f_k(x_k, u_k, w_k) \\ y_{k+1} &= h_{k+1}(x_{k+1}, v_{k+1}) \\ w_k &\sim (0, Q_k) \\ v_k &\sim (0, R_k) \end{aligned} \tag{15}$$

where x_{k+1} is the state vector, u_k is the control-input vector, f is the nonlinear state-transition function, y_{k+1} is the output vector, and w_k and v_k are the process and measurement noise, respectively. The matrices Q_k and R_k are the corresponding noise covariance matrices, and k denotes the time step in the discrete model.

The discrete EKF comprises two main stages: time update and measurement update. It can be applied to a nonlinear system through the following procedure.

I. Initialize the filter as follows:

$$\check{x}_0^+ = E(x_0)$$
$$P_0^+ = E[(x_0 - \check{x}_0^+)(x_0 - \check{x}_0^+)^T]$$

Here, $\check{x}_{k+1}^-$ is the a priori state estimate at step $k+1$, based on information available before the new measurement, whereas $\check{x}_{k+1}^+$ is the a posteriori estimate after incorporating y_{k+1}. Similarly, P_{k+1}^- and P_{k+1}^+ are the a priori and a posteriori estimation-error covariance matrices. The matrix F_k is the Jacobian of f with respect to X, and K_{k+1} is the Kalman gain that minimizes the error covariance. For $k = 1, 2, 3, \ldots, n$, the following stages are performed.

II. Derive the Jacobian matrices of the system equation:

$$F_k = \left.\frac{\partial f_k}{\partial X}\right|_{\check{\mathbf{x}}_\mathbf{k}^+}$$
$$L_k = \left.\frac{\partial f_k}{\partial w}\right|_{\check{\mathbf{x}}_\mathbf{k}^+}$$

The EKF time-update equations are

$$P_{k+1}^- = F_k P_k^+ F_k^T + L_k Q_k L_k^T \tag{16}$$
$$\check{x}_{k+1}^- = f_k(\check{x}_k^+, u_k, 0) \tag{17}$$

The Jacobian matrices of the output equation are obtained from

$$H_{k+1} = \left.\frac{\partial h_{k+1}}{\partial X}\right|_{\check{\mathbf{x}}_\mathbf{k+1}^-}$$
$$M_{k+1} = \left.\frac{\partial h_{k+1}}{\partial v}\right|_{\check{\mathbf{x}}_\mathbf{k+1}^-}$$

The final stage is the measurement update:

$$K_{k+1} = P_{k+1}^- H_{k+1}^T \left(H_{k+1} P_{k+1}^- H_{k+1}^T + M_{k+1} R_{k+1} M_{k+1}^T\right)^{-1} \tag{18}$$
$$\check{x}_{k+1}^+ = \check{x}_{k+1}^- + K_{k+1}\left[y_{k+1} - h_{k+1}(\check{x}_{k+1}^-, 0)\right] \tag{19}$$
$$P_{k+1}^+ = (I - K_{k+1} H_{k+1}) P_{k+1}^- \tag{20}$$

Because the EKF linearizes the system equations around each state estimate and retains only first-order Taylor-series terms, it may fail to represent strong system nonlinearities. The mean and covariance of the estimated states may then differ substantially from their actual values. The Unscented Kalman Filter (UKF), which approximates the state mean and covariance with higher-order accuracy, can address this limitation.

2) Unscented Kalman Filter (UKF) Algorithm: The UKF propagates the mean and covariance of the system state through nonlinear equations, capturing their behavior with accuracy up to the third order for certain classes of nonlinearities [14]. Using the system and measurement equations in (15), the complete procedure is described below.

I. Initialize the filter using the same initialization equations as the EKF.

II. Propagate the mean and covariance of the estimated state from one measurement instant to the next through the following time-update steps:

- Because the current best estimates of the mean and covariance of x_k are x_{k-1}^+ and P_{k-1}^+, respectively, the sigma points x_{k-1}^i are generated using the following equations and propagated through the nonlinear system from time $(k-1)$ to time k.

$$\check{x}_{k-1}^i = \check{x}_{k-1}^- + \tilde{x}^i \qquad i = 1, 2, ..., 2n$$
$$\tilde{x}^i = \left(\sqrt{nP_{k-1}^+}\right)_i^T \qquad i = 1, 2, ..., 2n$$
$$\tilde{x}^{n+i} = -\left(\sqrt{nP_{k-1}^+}\right)_i^T \qquad i = 1, 2, ..., 2n$$

Here, n is the number of system states, and $\sqrt{nP}$ denotes a matrix square root of nP such that $\left(\sqrt{nP}\right)^T \sqrt{nP} = nP$. The term $\left(\sqrt{nP_{k-1}^+}\right)_i^T$ denotes the ith row of the transposed matrix.

- Using the known nonlinear system equation f, propagate the $2n$ sigma points generated in the previous step through the system to obtain $\check{x}_k^i$:

$$\check{x}_k^i = f\left(\check{x}_{k-1}^i, u_k, t_k\right)$$

- Compute the a priori error-covariance matrix as follows:

$$P_k^- = \frac{1}{2n}\sum_{i=1}^{2n}\left(\check{x}_k^i - \check{x}_k^-\right)\left(\check{x}_k^i - \check{x}_k^-\right)^T + Q_{k-1}$$

- For the measurement update, select new sigma points from the updated values of $\check{x}_k^-$ and P_k^-:

$$\begin{aligned}\check{x}_k^i &= \check{x}_k^- + \tilde{x}^i \qquad i = 1,2,...,2n\\ \tilde{x}^i &= \left(\sqrt{nP_k^-}\right)_i^T \qquad i = 1,2,...,2n\\ \tilde{x}^{(n+i)} &= -\left(\sqrt{nP_k^-}\right)_i^T \qquad i = 1,2,...,2n\end{aligned} \tag{21}$$

- Transform the sigma points $\check{x}_k^i$ into $\check{y}_k^i$ using the known nonlinear output equation h:

$$\check{y}_k^i = h(\check{x}_k^i, t_k)$$

- Obtain the predicted measurement at time step k as follows:

$$\check{y}_k^i = \frac{1}{2n}\sum_{i=1}^{2n}\check{y}_k^i$$

- Calculate the predicted-measurement covariance:

$$P_y = \frac{1}{2n}\sum_{i=1}^{2n}\left(\check{y}_k^i - \check{y}_k\right)\left(\check{y}_k^i - \check{y}_k\right)^T + R_k$$

- Calculate the cross-covariance between $\check{x}_k^-$ and $\check{y}_k$:

$$P_{xy} = \frac{1}{2n}\sum_{i=1}^{2n}\left(\check{x}_k^i - \check{x}_k^-\right)\left(\check{y}_k^i - \check{y}_k\right)^T$$

- Finally, perform the measurement update:

$$\begin{aligned}K_k &= P_{xy}P_y^{-1}\\ \check{x}_k^+ &= \check{x}_k^- + K_k(y_k - \check{y}_k)\\ P_k^+ &= P_k^- - K_k P_y K_k^T\end{aligned} \tag{22}$$

Unlike the first-order approximation used by the EKF, the UKF can approximate the mean and covariance of the system state with higher-order accuracy. It generally represents the mean and covariance of strongly nonlinear systems in the presence of noise more accurately than the EKF, although it requires greater computational effort.

D. Summary of Methods

Table I lists several state-estimation methods identified in the literature review.

TABLE I
SUMMARY OF STATE-ESTIMATION METHODS

Weighted least absolute value
Constrained least squares
Pseudoinverse-matrix methods
Least median of squares
Hybrid methods
Artificial neural networks
Support vector machines
Fuzzy logic

III. Error Statistics and Bad-Data Processing

Even when measurements are acquired with great care, random disturbances inevitably introduce noise into power-system data-acquisition processes. Although sometimes small, these distortions may compromise the quality of the physical quantities used in subsequent studies. Repeated measurements of the same quantity under controlled conditions nevertheless reveal statistical properties that can be used to infer its true value [6].

The accuracy and reliability of state estimation are directly related to measurement quality. Measurements may originate from analog or digital devices and are subject both to instrument-accuracy limitations and to distortions introduced by communication channels. If these errors produce bad data that are not detected and identified, the estimated system state may be incorrect [1, 12]. State estimators generally include preprocessing to identify obvious inconsistencies, but this is insufficient in sparsely instrumented areas. WLS estimators therefore employ post-estimation statistical analysis of the residuals to detect and identify bad data, as indicated in Fig. 4.

Effective error characterization and management require a detailed understanding of error sources. Modeling errors arise from inaccurate system representations, such as incorrect transmission-line parameters or erroneous indications of circuit-breaker status. Data errors may result from incorrectly specified connections between current transformers and transducers, as well as wiring or contact-assignment mistakes. Transducer errors may be systematic or random and affect the fidelity of measured signals. Finally, sampling errors arise from the asynchronous nature of SCADA systems, which do not guarantee simultaneous measurements or uniform reporting rates and may therefore distort the representation of dynamic system behavior [8].

Understanding the statistical nature of measurement errors and applying appropriate detection and correction procedures are fundamental to the design of robust, reliable estimators. These practices provide safer and more accurate support for real-time operation and decision-making in control centers.

The ability to detect and identify measurement errors in state estimation depends critically on redundancy in the available data set. Without sufficient redundancy, error identification is impossible regardless of the method used [1].

Measurements can be divided into two mutually exclusive groups: critical and noncritical (redundant) measurements. A measurement is critical if its removal makes the system unobservable. Its associated residual is always zero, making errors in that measurement undetectable [5]. Noncritical measurements may permit error identification through statistical residual analysis, although noncriticality alone does not guarantee that an error can be detected [1].

Bad-data analysis generally comprises detection, identification, and removal. Residuals are the primary quantities used to determine whether a measurement should be considered suspicious. According to [5], two tests are commonly used:

the chi-square (χ^2) test for bad-data detection and the largest normalized residual (r^N) test for bad-data identification.

Plotting measured values against their relative frequency of occurrence produces a histogram. As the number of measurements increases, a continuous curve can be fitted to this histogram. The most common model is the bell-shaped normal, or Gaussian, probability density function $f(z)$ [6], given by

$$f(z) = PDF = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(z-u)^2/\sigma^2} \tag{23}$$

The sum of the squares of independent standard normal random variables follows a chi-square distribution $\chi^2_{(K)}$. Under the assumptions that measurement errors are independent and normally distributed, the objective function $J(x)$ in (7) can therefore be shown to follow a chi-square distribution [8]. The parameter K, which denotes the number of degrees of freedom, is defined as

$$K = N_m - N_s = N_m - (2n - 1)$$

where N_m is the number of measurements, N_s is the number of states, and n is the number of buses in the network.

Bad-data detection using this test proceeds as follows [1]:

- Solve the WLS estimation problem and calculate the objective function in (7). The measurement weight is commonly defined as $W_i = R_i^{-1} = (\sigma_i^2)^{-1}$.
- Select a probability level and determine the chi-square threshold $\chi^2_{(K,p)}$ for K degrees of freedom, where $p = \Pr[J(\check{x}) \leq \chi^2_{(K,p)}]$.
- If $J(\check{x}) \geq \chi^2_{(K,p)}$, bad data are suspected. Otherwise, there is no statistical evidence of bad data at the selected probability level.

Bad-data identification determines which measurement or measurements are erroneous after a detection method has indicated that bad data may be present. The normalized-residual test is the best-known identification method because it is easy to implement and generally performs satisfactorily, except in the presence of interacting or conforming multiple bad data.

The method assumes that normalized residuals follow a standard normal distribution with zero mean and unit variance [6, 1]. Their formulation is shown in (24).

$$r_i^N = \frac{r_i}{\sqrt{\Omega_{ii}}} = \frac{e_i - \check{e}_i}{\sqrt{\Omega_{ii}}} \tag{24}$$

If there is a single noncritical bad measurement, the largest normalized residual can be shown to correspond to that measurement. The normalized-residual test for identifying a single bad measurement or multiple noninteracting bad measurements proceeds as follows [5, 1]:

1) Solve the WLS estimation problem and obtain the residual-vector elements r_i.
2) Calculate the normalized residuals r_i^N.
3) Determine the index k for which $|r_k^N|$ is the largest among all $|r_i^N|$.
4) If $|r_k^N| > c$, classify measurement k as bad. The parameter c is a suitably selected identification threshold, commonly 3.
5) Remove the bad measurement k and repeat the procedure.

The residual covariance matrix Ω, whose diagonal elements are Ω_{ii}, is obtained from

$$\mathbf{\Omega} = \mathbf{R'} = \mathbf{R} - \left(\mathbf{HG^{-1}H^T}\right), \quad \mathbf{R} = \mathbf{W^{-1}} \tag{25}$$

IV. Conclusions

- State estimation is essential for real-time power-system monitoring because it determines the internal operating state from a limited and noisy set of measurements. By smoothing random errors, detecting bad data, and compensating for missing information, it creates a reliable database for safe and efficient grid operation and supports critical applications such as economic dispatch, security analysis, and voltage regulation. Its use in modern control centers is therefore fundamental to reliable, efficient, and resilient operation under disturbances and abnormal conditions.
- PMUs have substantially advanced state estimation by providing high-rate, high-accuracy synchronized measurements that improve estimation quality relative to conventional SCADA-only systems. They enable faster and more accurate responses to system events, increase observability, facilitate the detection of abnormal dynamics, and provide information for advanced protection and control strategies. Strategic PMU deployment improves static estimation and supports the development of more robust and adaptive dynamic and hybrid schemes.
- Bad measurements must be detected and identified before operational actions are taken because even small disturbances or communication failures can degrade estimation quality and lead to incorrect decisions. Error identification depends critically on measurement redundancy; without adequate redundancy, some errors cannot be detected. Statistical procedures such as the chi-square test for detection and the largest normalized residual test for identification help ensure that only reliable data are used in the estimation process.
- A wide range of state-estimation methods is available, from conventional WLS to robust estimators and dynamic techniques based on Kalman filters. Each involves tradeoffs: methods offering greater accuracy or robustness to outliers may require more computation or more complex configuration. Current research increasingly explores artificial intelligence and hybrid techniques that combine multiple measurement sources to produce faster, more adaptive, and fault-tolerant estimators.

V. Recommendations

- To improve both the quality and speed of power-system state estimation, future work should strengthen the integration and synchronization of SCADA and PMU

data. Artificial-intelligence or machine-learning techniques may be used to adjust measurement weights dynamically according to confidence in each sensor. Models trained on historical operating data could predict instrumentation errors and update each sensor's contribution in real time, improving operating-point accuracy and the detection of transients and anomalies.
- Because installing a PMU at every bus is impractical, PMU placement should be optimized to achieve maximum observability with the fewest units. Optimization algorithms should account for budgetary and topological constraints when selecting buses, seek broad system coverage, and minimize the occurrence of critical measurements whose errors cannot be detected. This approach would improve the reliability and resilience of state estimation.

VI. References

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